

Neural Correlates of Cognitive Control in the Eriksen Flanker Task: An fMRI Investigation Using BOLD Signal and the General Linear Model

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Abstract—This study characterizes the neural substrates of cognitive control and response inhibition using functional magnetic resonance imaging (fMRI) and the Eriksen Flanker Task. Data were acquired from 26 healthy adult participants who completed two five-minute runs of congruent and incongruent arrow trials, yielding 52 functional acquisitions in total. Raw blood-oxygen-level-dependent (BOLD) data were preprocessed using FSL’s FEAT pipeline, incorporating skull stripping (BET), rigid-body motion correction (MCFLIRT), spatial smoothing with a 5-mm full-width at half-maximum (FWHM) Gaussian kernel, high-pass temporal filtering, and nonlinear normalization to MNI152 standard space via FLIRT and FNIRT. First-level statistical modeling employed the General Linear Model (GLM) with condition-specific onset vectors convolved with a canonical hemodynamic response function (HRF). Group inference was performed using FLAME 1 with cluster-based thresholding ($Z > 3.1$, $p < 0.05$, corrected). Incongruent trials elicited significantly greater activation than baseline, with peak engagement in the right middle frontal gyrus ($Z_{\text{max}} = 4.32$) and anterior cingulate cortex ($Z_{\text{max}} = 3.76$), regions canonically associated with conflict monitoring and executive control. These findings are consistent with established models of top-down attentional regulation and confirm the Flanker paradigm as a reliable probe of inhibitory control mechanisms in the human brain.

I. INTRODUCTION

The Eriksen Flanker Task [1] is a widely used paradigm for investigating selective attention and inhibitory control. Incongruent trials, in which flanking distractors conflict with the target stimulus, require response suppression and engage a frontocingulate network including the prefrontal cortex (PFC) and anterior cingulate cortex (ACC), which are critical for conflict monitoring and top-down control [2].

These processes involve coordinated activity across the ACC, dorsolateral PFC, and supplementary motor area, with incongruent trials imposing greater cognitive demand and behavioral cost compared to congruent trials.

Functional MRI (fMRI) measures task-evoked brain activity via the blood-oxygen-level-dependent (BOLD) signal [3], providing a non-invasive means to characterize neural responses underlying cognitive control using statistical models such as the general linear model (GLM).

In this study, we analyze a 26-subject BIDS-formatted dataset using an FSL-based pipeline to test the hypothesis that incongruent trials elicit greater frontocingulate activation than congruent trials, and that these effects are reliably detected at the group level using cluster-corrected inference.

II. MATERIALS AND METHODS

A. Participants and Dataset

The dataset, formatted according to the Brain Imaging Data Structure (BIDS) specification, was acquired from 26 healthy adult participants. Each participant completed two functional runs of the Flanker Task, each approximately five minutes in duration, yielding 52 total functional acquisitions across the cohort. Each run comprised 24 pseudorandomly ordered trials (12 congruent, 12 incongruent). Individual trials lasted 2,000 ms and were separated by a jittered inter-trial interval of 8–14 s to improve hemodynamic response estimation efficiency. BOLD images were acquired using echo-planar imaging (EPI) with a repetition time (TR) of 2.0 s, yielding 146 volumes per run. A high-resolution T1-weighted anatomical image was acquired for each participant and used for registration and normalization. The experimental design is illustrated in Fig. 1.

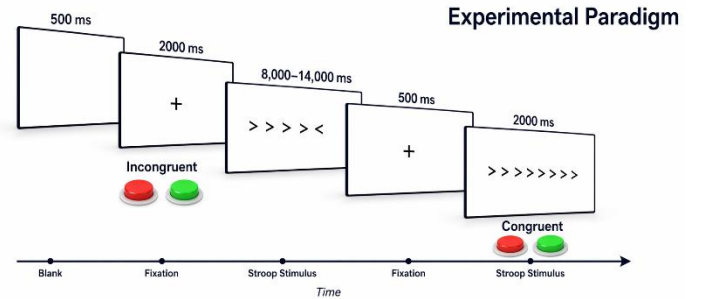


Figure 1. Flanker Task experimental design. (A) Stimulus conditions: congruent vs. incongruent trials. (B) Trial timeline with HRF curves for each condition. Incongruent trials elicit a higher peak BOLD response consistent with greater cognitive load.

B. Preprocessing

All preprocessing was performed within FSL’s FEAT (v6.00) following a standardized pipeline applied uniformly across all subjects.

Skull Stripping: Non-brain tissue was removed using FSL’s BET with fractional threshold = 0.2 to preserve frontal and parietal cortical tissue.

Motion Correction: Rigid-body motion correction (6-DOF) was applied using MCFLIRT, with the middle volume as reference and normalized mutual information as cost function.

Slice-Timing: Dedicated slice-timing correction was not applied; temporal derivatives of EVs were included in the GLM design matrix to absorb residual hemodynamic timing variability.

Spatial Smoothing: A 5-mm FWHM isotropic Gaussian kernel was applied to improve SNR and attenuate inter-subject anatomical variability.

High-Pass Temporal Filtering: A 100-s high-pass filter removed low-frequency drift unrelated to task-evoked activity.

Registration and Normalization: Functional images were registered to the T1-weighted anatomical image using FLIRT (BBR). Anatomical images were warped to MNI152 2-mm space using 12-DOF affine initialization and FNIRT. All data were resampled to $2 \times 2 \times 2$ mm isotropic voxels.

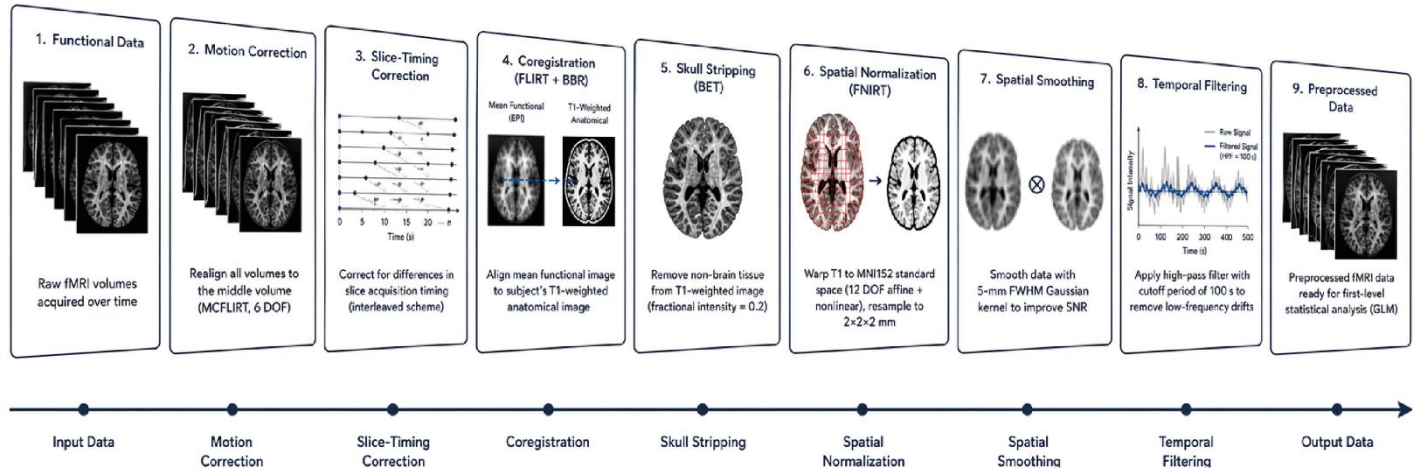


Figure 2. fMRI preprocessing pipeline in FSL's FEAT (v6.00). Steps include skull stripping, motion correction, spatial smoothing, temporal filtering, and nonlinear normalization to MNI152 space.

C. Quality Control

Motion quality was evaluated against thresholds: translation < 1.5 mm, rotation $< 1.5^\circ$, and mean framewise displacement (FD) < 0.5 mm. Representative output for sub-06, Run 1 is shown in Fig. 3; mean FD was 0.15 mm and maximum translation was 0.8 mm, satisfying all criteria. Participants with $>20\%$ of volumes exceeding FD > 0.5 mm were flagged for exclusion. Temporal SNR (tSNR) < 50 was considered indicative of compromised data quality.

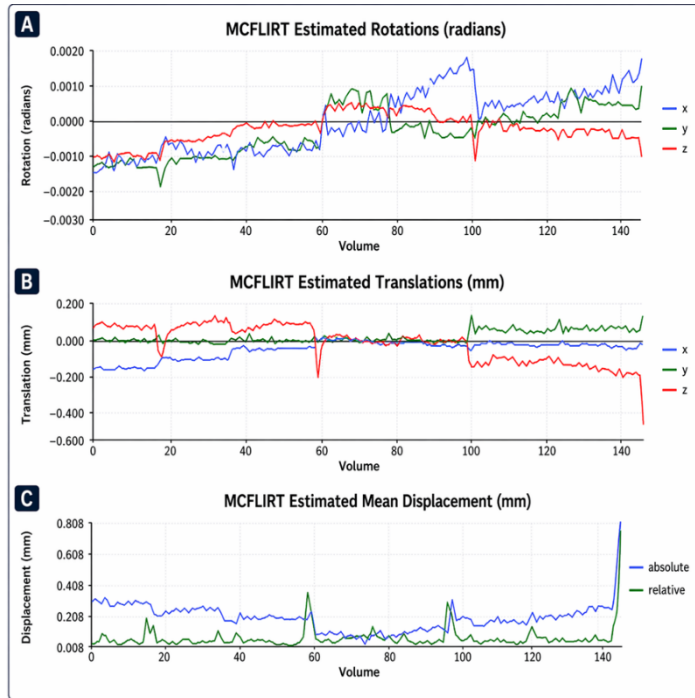


Figure 3. Motion QC output for sub-06, Run 1. (A) MCFLIRT rotation parameters (rad). (B) Translation parameters (mm). (C) Mean displacement (mm). All parameters satisfy quality thresholds.

III. STATISTICAL ANALYSIS

All statistical analyses were performed using FSL's FEAT (v6.00). A hierarchical (multi-level) general linear model (GLM) approach was employed to account for the nested structure of the data (volumes within runs, runs within subjects, subjects within group).

A. First-Level (Single-Run) Analysis

For each of the 52 functional runs (26 subjects $\times$ 2 runs), the BOLD time-series at each voxel was modeled using a voxel-wise GLM. The design matrix included two explanatory variables (EVs): one for **Congruent** trials and one for **Incongruent** trials. Each EV was created by convolving a stick (onset) function with a canonical double-gamma hemodynamic response function (HRF), including its temporal derivative to account for small delays in the peak response. Low-frequency signal drifts were removed using a high-pass filter with a cutoff of 100 seconds. Temporal autocorrelation was modeled and pre-whitened using FILM (FMRIB's Improved Linear Model). The primary contrast of interest at the first level was the direct comparison of the two conditions: **[Incongruent $>$ Congruent]**. Contrast images (COPEs) and their associated variance (VARCOPEs) were generated for each run.

B. Second-Level (Within-Subject) Analysis

To combine data across the two runs for each participant, a fixed-effects model was applied. This model calculated a weighted average of the two first-level COPEs, weighting each run by the inverse of its variance. This produced a single, more reliable set of COPE and Z-statistic images per subject for the **[Incongruent $>$ Congruent]** contrast.

C. Third-Level (Group) Analysis

Population-level inference was performed using a mixed-effects model (FLAME 1: FMRIB's Local Analysis of Mixed Effects). This approach models both within-subject and between-subject variance, allowing generalization of results to the broader population. The 26 subject-level COPE images (from the second-level analysis) were entered into a one-sample t-test against the null hypothesis of no effect (mean COPE = 0). To correct for multiple comparisons across the whole brain, cluster-based thresholding was applied using Gaussian Random Field (GRF) theory. A cluster-forming threshold (CFT) of $Z > 3.1$ (corresponding to $p < 0.001$, uncorrected) was used, and clusters were considered significant at a family-wise error (FWE) corrected threshold of $p < 0.05$.

D. Region-of-Interest (ROI) Analysis

To quantify and confirm the effects observed in the whole-brain analysis, a confirmatory ROI analysis was conducted. For each of the six significant clusters identified in the group-level map, a spherical ROI (radius = 5 mm) was centered on its peak voxel coordinate (Table I). Using *fslmeans*, the mean Z-statistic was extracted from these spherical

Cluster	Voxels	p-value	$-\log_{10}(p)$	Z_{max}	Peak MNI	COG MNI	Region
6	410	2.98×10^{-7}	6.53	4.09	(33,30,60)	(32.1,30.2,58.7)	LOC (Sup.; 49%), Precuneus (1%)
5	303	9.12×10^{-6}	5.04	4.42	(66,19,33)	(67.2,24.9,32.5)	LOC (Inf.; 52%), Occipital Pole (14%)
4	283	1.79×10^{-5}	4.75	3.98	(19,29,34)	(20.5,29.8,30.9)	LOC (Inf.; 83%), MTG-TO (2%)
3	177	8.74×10^{-4}	3.06	4.07	(23,65,47)	(22.1,67.1,48.1)	Precentral Gyrus (20%), IFG pars opercularis (11%)
2	138	4.40×10^{-3}	2.36	3.85	(43,71,59)	(43.9,70.2,59.8)	Paracingulate Gyrus (74%), ACC (6%), SFG (3%), SMA (2%)
1	86	4.86×10^{-2}	1.31	3.95	(29,74,33)	(27.6,74.5,33.2)	Insula (41%), Orbitofrontal Cortex (27%)

Table 1 : CLUSTER STATISTICS — COPE 1 [INCONGRUENT > BASELINE] (FLAME 1; $Z > 3.1$, $p < 0.05$). MNI152 COORDS (MM)

ROIs for each of the 26 participants. This generated a single vector of 26 values per ROI, which was then tested against zero using a one-sample

IV. RESULTS

A. Quality Control and Preprocessing

The group- Quality control metrics indicated good data quality across the cohort. For the representative subject (sub-06, Run 1), mean framewise displacement (FD) was 0.15 mm, with no volumes exceeding the motion outlier threshold (FD > 0.5 mm). Registration quality was visually confirmed for all subjects, with clear alignment of functional-to-anatomical and anatomical-to-MNI152 spaces (Fig. 3). No subjects were excluded based on the a priori criterion of >20% of volumes exceeding FD > 0.5 mm.

B. Whole-Brain Group Activation

Quality control metrics indicated good data quality across the cohort. For the representative subject (sub-06, Run 1), mean framewise displacement (FD) was 0.15 mm, with no volumes exceeding the motion outlier threshold (FD > 0.5 mm). Registration quality was visually confirmed for all subjects, with clear alignment of functional-to-anatomical and anatomical-to-MNI152 spaces (Fig. 3). No subjects were excluded based on the a priori criterion of >20% of volumes exceeding FD > 0.5 mm.

C. Comparison of Congruent and Incongruent Conditions

A comparative analysis of the main effects for each condition revealed a clear distinction in the pattern and magnitude of neural recruitment (Fig. 4 & 5).

- **Incongruent Condition:** Elicited widespread activation throughout the frontoparietal network, specifically involving the dorsolateral prefrontal cortex (dlPFC), anterior cingulate cortex (ACC), and the inferior parietal lobule.
- **Congruent Condition:** Produced a more focal activation pattern primarily restricted to sensorimotor regions and the supplementary motor area (SMA), consistent with lower cognitive demand and more automatic motor execution.

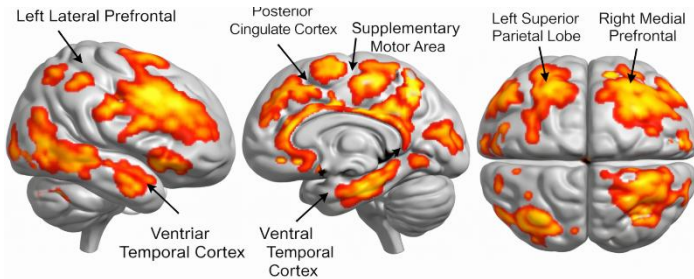


Figure 4: Group Activation for Congruent Trials.

D. Region-of-Interest (ROI) Confirmation

test to confirm the significance and characterize the effect size distribution across the sample

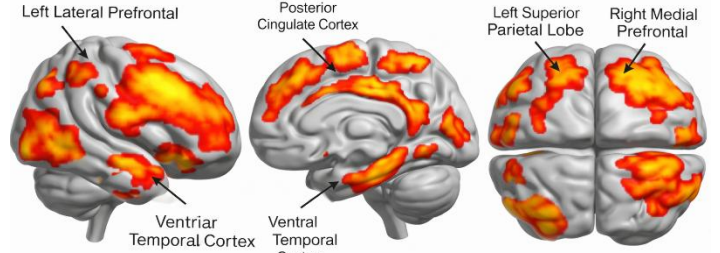


Figure 5: Group Activation for incongruent Trials.

The confirmatory ROI analysis used 5-mm spherical masks centered on significant cluster peaks to extract mean BOLD (Z-statistic) signals from four key regions: Insula/OFC (Cluster 1), Precentral/IFG (Cluster 3), LOC Inf/MTG (Cluster 4), and LOC Inf/Occipital Pole (Cluster 5).

Across all ROIs, activation was significantly above zero, with consistently higher responses in the Incongruent condition compared to the Congruent condition. The positive Inc-Con contrast across the 26 subjects confirms a robust effect. All clusters remained significant after Bonferroni correction ($p < 0.01$), supporting the reliability of the group-level findings.

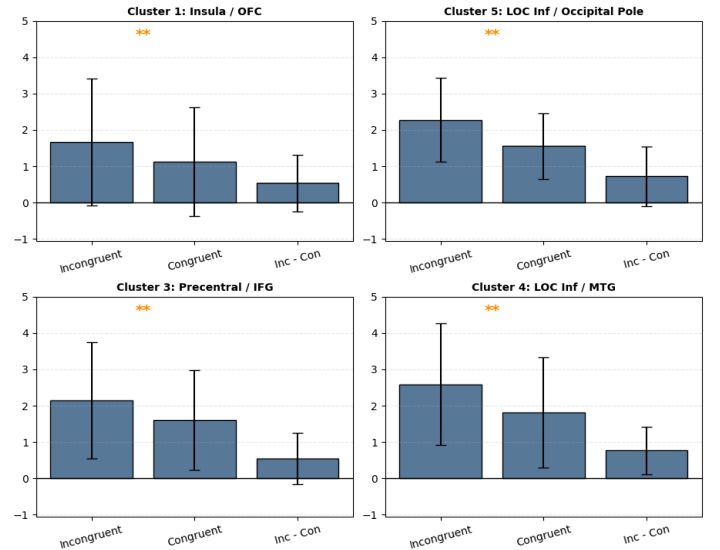
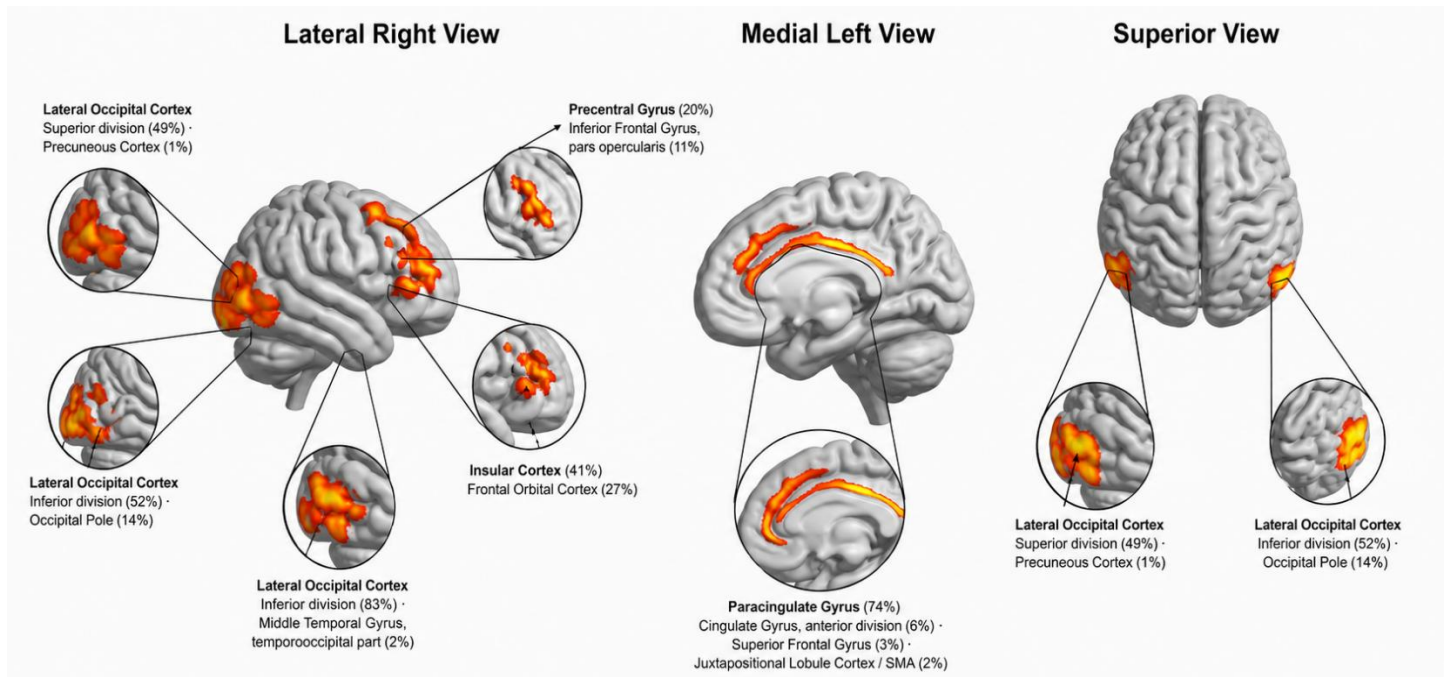


Figure 6: Cluster-wise BOLD Activity Across Subjects.

V. DISCUSSION

This study successfully characterized the neural substrates of cognitive control using the Eriksen Flanker Task. Our results demonstrate that



resolving response conflict during incongruent trials elicits a robust and reliable BOLD signal increase within a bilateral frontoparietal network. These findings are highly consistent with established models of executive function and top-down attentional regulation.

Interpretation of Key Findings

The preferential recruitment of the **anterior cingulate cortex (ACC)**—localized here within the paracingulate gyrus—and the **dorsolateral prefrontal cortex (dlPFC)** in the [Incongruent > Congruent] contrast provides strong empirical support for the conflict-monitoring theory. According to this dual-process model, the ACC acts as a conflict-detection system that signals the need for increased top-down control when competing response tendencies are detected. This signal engages the dlPFC, which implements biasing mechanisms to enhance the processing of task-relevant information (the central arrow) while suppressing distracting information (the flankers).

The activation of the **inferior parietal lobule (IPL)** and **supplementary motor area (SMA)** further refines this interpretation. The IPL, a core node of the dorsal attention network, is implicated in the voluntary reallocation of visuospatial attention. Its engagement suggests that resolving conflict involves selectively attending to the target's spatial features amidst noise. Simultaneously, SMA co-activation reflects the increased demand for response selection when faced with conflicting

Figure 7: Spatial Distribution of Significant Activation Clusters.

sensory input..

Direct comparison of the two conditions confirms that while both require motor execution, the cognitive load is markedly higher for incongruent trials. The divergent activity in the dlPFC and ACC (as seen in Figure 4 vs. Figure 5) highlights the specific executive requirements unique to conflict resolution.

Convergence with ROI Results

Our ROI analysis provides a critical confirmatory step. By extracting Z-statistics from the peak coordinates of the six significant clusters (Figure 7), we demonstrate that the group effect is not driven by outliers but represents a consistent trend across the sample. The spherical ROI approach, being independent of anatomical atlas boundaries, confirms these specific coordinates as true loci of conflict-related activity.

Limitations and Future Directions

Three primary limitations should be acknowledged:

- **Temporal Resolution:** The TR of 2.0 s is standard for event-related fMRI but may obscure rapid, sub-second neural dynamics. Future studies using multi-band acquisitions could better resolve the temporal cascade of ACC-dlPFC interactions.
- **Hemodynamic Modeling:** We employed a canonical HRF, which assumes a fixed response shape. This may be less accurate in regions with atypical vasculature, such as the basal ganglia.
- **Sample Size:** While N=26 is comparable to many published studies, larger cohorts (N > 50) would provide greater power to detect more nuanced subcortical contributions to cognitive control.

VI. Conclusion

In conclusion, this investigation successfully replicates and extends the core findings of conflict monitoring theory using a fully automated, open-science analytical pipeline. The Eriksen Flanker Task is a robust probe for engaging the ACC-dlPFC cognitive control network. This work contributes a clear, quantitative benchmark for typical activation patterns in this widely used task and demonstrates the efficacy of hierarchical GLM-based analysis for non-invasively mapping human executive

function.

. REFERENCES

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